Name: _____

WORKSHEET

# Limits

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 16 | 5 | 16 | 13 | | 9 | 16 | 8 | | 8 | 2 | 16 | | 17 | 16 | 10 | 13 | | |
| 4 | 17 | | | | | 14 | 8 | 13 | 12 | 11 | 12 | 6 | 7 | | 4 | 1 | 8 | | |
| 7 | 16 | 8 | | | | 12 | 6 | | | 3 | 4 | 1 | 13 | | 15 | 10 | 3 | | |

*George Herman 'Babe' Ruth*

## Questions

Find the following limits.

**1** $\lim_{x \to 0} (2x + 11)$

**2** $\lim_{x \to 3} (20 - 4x)$

**3** $\lim_{x \to -1} (3x^2 - x + 5)$

**4** $\lim_{x \to 2} \dfrac{(6x^2 + 3x + 10)}{4x}$

**5** $\lim_{x \to 0} 4x^2 - x - 7$

**6** $\lim_{x \to \infty} (5x - 2)$

**7** $\lim_{x \to 5} \dfrac{7 + x}{x^2 - 1}$

**8** $\lim_{x \to \infty} \dfrac{3}{x}$

**9** $\lim_{x \to \infty} \dfrac{-2x + 7}{x}$

**10** $\lim_{x \to \infty} \dfrac{6x^2 + 3x - 1}{2x^2 + 7x + 3}$

**11** $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$

**12** $\lim_{x \to 4} \dfrac{6x - 24}{x - 4}$

**13** $\lim_{x \to -1} \dfrac{x^2 - 2x - 3}{x + 1}$

**14** $\lim_{x \to \frac{1}{3}} \dfrac{6x^2 + 13x - 5}{9x - 3}$

**15** $\lim_{x \to 5} \dfrac{15 - 3x}{3x^2 - 17x + 10}$

**16** $\lim_{x \to 2} \dfrac{2x^2 - x - 6}{x^2 + x - 6}$

**17** $\lim_{x \to -3} \dfrac{x^2 + 2x - 3}{x^2 + 4x + 3}$

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## Solutions

**A** $3$

**E** $1\frac{2}{5}$

**F** $2$

**G** $\frac{1}{2}$

**H** $8$

**I** $6$

**K** $4$

**L** $-2$

**N** $\infty$

**O** $5$

**R** $-4$

**S** $1\frac{8}{9}$

**T** $0$

**U** $11$

**V** $-7$

**W** $-\frac{3}{13}$

**Y** $9$

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